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Title

**Assessment of Alzheimer's Disease
Progression Using a
Comorbidity-Aware Meta-Bayesian
Framework with
Resting-State fMRI**

REVIEW REPORT – II

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Abstract

Alzheimer’s disease (AD) develops over many years, moving from cognitively normal ageing through subjective memory complaints and successive stages of mild cognitive impairment (MCI) before dementia. Detecting where a person sits on this continuum, rather than only separating AD from healthy controls, is what makes early intervention possible. This project stages AD progression across six ordered classes (CN $\rightarrow$ SMC $\rightarrow$ EMCI $\rightarrow$ MCI $\rightarrow$ LMCI $\rightarrow$ AD) from resting-state functional MRI (rs-fMRI) of the Alzheimer’s Disease Neuroimaging Initiative (ADNI). The planned model combines a multi-attention transformer, meta-learning and comorbidity fusion with a Bayesian ordinal classifier that reports uncertainty with each prediction.

This report covers the data foundation built since the first review. The ADNI data has no T1-weighted anatomical image and no recoverable slice-timing metadata, so every processing step was designed around a BOLD-only dataset. 175 acquisitions were audited and 29 missing scans were recovered from raw DICOM. Two acquisitions were excluded with documented reasons. A fixed pipeline was then applied: motion correction, direct EPI-to-MNI registration, nuisance regression and band-pass filtering. It produced 165 preprocessed scans from 127 subjects. The 8 remaining scans were rejected because their TR of 6.02 s makes the 0.01–0.10 Hz band invalid. Each scan was parcellated with the Brainnetome-246 atlas into ALFF, ReHo, functional connectivity (FC) and degree-centrality (DC) biomarkers. These were then normalised within each subject, so no information leaks across subjects. All 165 scans passed the independent consistency checks. Recomputing DC from the stored FC matrices matched exactly (difference 0.0). The dataset-scale code reproduced the validated pilot bit for bit. One nominal feature, FC strength, turned out to be exactly DC/245 and was removed. Each scan is therefore described by a 246×3 regional matrix and a 246×246 Fisher- z connectivity matrix. The modelling stages are specified and scheduled. Their evaluation will use subject-grouped, stratified cross-validation.

Keywords: Alzheimer’s disease; resting-state fMRI; ADNI; Brainnetome atlas; ALFF; ReHo; functional connectivity; ordinal classification; Bayesian deep learning; meta-learning

1. Introduction

Alzheimer’s disease is the most common cause of dementia and is a growing burden on ageing populations [1, 2]. Its pathology starts years before clinical symptoms appear. It is now viewed as a continuum: from cognitively normal (CN) ageing, to subjective cognitive decline or subjective memory concern (SMC) [3], to early, intermediate and late mild cognitive impairment (EMCI, MCI, LMCI) [4], and finally to dementia due to AD. Intervention is most promising in the early stages. So the clinically useful question is not only “does this person have AD?” but “where on the continuum is this person?”

Resting-state functional MRI (rs-fMRI) measures spontaneous low-frequency fluctuations of the blood-oxygen-level-dependent (BOLD) signal while the subject lies at rest [5]. It needs no task performance, which makes it practical for cognitively impaired patients. AD-related disruption of intrinsic networks, notably the default-mode network, was reported early on [6]. Regional measures such as the amplitude of low-frequency fluctuation (ALFF) [7] and regional homogeneity (ReHo) [8], together with network measures such as functional connectivity (FC) and degree centrality (DC) [9], capture complementary aspects of this disruption.

Imaging alone ignores factors that change an individual’s risk and trajectory. Examples are the APOE $\epsilon 4$ allele [10], age, sex and vascular or metabolic comorbidities [2]. AD stages are also *ordered*: confusing CN with SMC is a smaller error than confusing CN with AD. Treating staging as flat multi-class classification ignores this structure. These two observations motivate the project. It aims to build a *comorbidity-aware*, *ordinal* and *uncertainty-reporting* staging model on top of a carefully validated rs-fMRI feature set.

1.1 Scope of this Review

The first review fixed the problem, the six-class progression ordering and the overall architecture. Since then the work has concentrated on the data foundation that every later stage depends on. That covers dataset audit and recovery, preprocessing of all six groups, quality control (QC), Brainnetome-246 parcellation, biomarker generation, independent validation and leakage-free within-subject normalisation. All of these are complete for the full dataset and are reported here with their verification evidence. The learning stages (multi-attention transformer, meta-learning, comorbidity fusion and Bayesian ordinal classifier) are specified in Chapter 4. They are scheduled in Section 6.8 but are not yet implemented, and no model results are claimed in this

report.

1.2 Objectives

1. To curate an analysis-ready ADNI rs-fMRI cohort covering six diagnostic stages, with every exclusion, recovery and metadata limitation documented rather than assumed.
2. To design and implement a reproducible, BOLD-only preprocessing pipeline with per-acquisition quality control and full provenance.
3. To extract regional (ALFF, ReHo, DC) and network (FC) biomarkers on the Brainnetome-246 atlas and to verify them independently of the code that generated them.
4. To build a leakage-free, model-ready feature representation for every acquisition.
5. To develop a comorbidity-aware Meta-Bayesian model, combining a multi-attention transformer, meta-learning, comorbidity fusion and a Bayesian ordinal classifier, that stages AD progression with calibrated uncertainty.
6. To evaluate the model under subject-grouped cross-validation with ordinal and calibration metrics, and to explain its predictions at the level of brain regions.

Objectives 1–4 are complete; objectives 5–6 form the remaining work.

1.3 Organisation of the Report

Chapter 2 answers the queries raised in the first review. Chapter 3 states the problem formally. Chapter 4 presents the architecture and modules. Chapter 5 describes the implementation, and Chapter 6 reports and discusses the results obtained so far, together with the limitations and the plan for the remaining work.

2. First Review – Queries and Answers

- **Query 1:** [Query raised by the panel in Review I.]
Answer: [Answer and the section of this report where it is addressed.]
- **Query 2:** [Query raised by the panel in Review I.]
Answer: [Answer.]
- **Query 3:** [Query raised by the panel in Review I.]
Answer: [Answer.]

2.1 Design Revisions Since Review I

Verifying the data turned up several facts that change or tighten the Review-I design. They are listed here so that the panel can see what changed and why.

- R1. Regional feature matrix is 246×3 , not 246×4 .** The Review-I design used four regional biomarkers (ALFF, ReHo, DC, FC strength). FC strength is the *mean* of a region’s off-diagonal connectivity and DC is the *sum* of the same entries, so $FCS_i = DC_i/245$ exactly (measured Pearson $r = 1.0$; maximum deviation 6.9×10^{-18} over 40,590 region values). FC strength carries no independent information and was removed from the feature set (Section 5.5).
- R2. Effective sample size is 127 subjects, not 165 scans.** 96 subjects have one scan, 25 have two, 5 have three and 1 has four. MCI is the largest class by scans (32) but the smallest by subjects (14). All validation splits will therefore be grouped by subject (Section 4.3.6).
- R3. No AD acquisition is unrecoverable.** An earlier working note listed 6 of 25 AD scans as unrecoverable. A re-audit located all 6 as valid, checksummed NIfTI files, and all 25 AD scans were processed.
- R4. Progression ordering is held fixed** as $CN \rightarrow SMC \rightarrow EMCI \rightarrow MCI \rightarrow LMCI \rightarrow AD$. The ordinal loss depends on it.
- R5. Hypertension and diabetes are not directly available.** The supplied ADNI medical-history tables carry only broad categories (cardiovascular, endocrine-metabolic). The comorbidity module will use these categories under their real names instead of relabelling them as specific diseases.

3. Problem Statement

Task. Given the resting-state fMRI acquisition(s) and the clinical and comorbidity record of a subject, estimate the subject’s stage on the ordered scale

$$y \in \{0: \text{CN}, 1: \text{SMC}, 2: \text{EMCI}, 3: \text{MCI}, 4: \text{LMCI}, 5: \text{AD}\},$$

and report a predictive distribution $p(y \mid \mathbf{X}, \mathbf{c})$ whose spread reflects the model’s uncertainty.

Inputs.

- A 4-D BOLD time series per acquisition (ADNI, Philips scanners, mostly TR ≈ 3.0 s, 140 volumes, ≈ 3.3 mm voxels).
- From it, after this review’s work: a regional feature matrix $\mathbf{R} \in \mathbb{R}^{246 \times 3}$ (mALFF, mReHo, DC_z) and a connectivity matrix $\mathbf{Z} \in \mathbb{R}^{246 \times 246}$ (Fisher- z FC).
- A comorbidity vector $\mathbf{c}$: age, sex, APOE genotype, and clinical and medical-history variables where available.

Outputs. A stage estimate $\hat{y}$, the full distribution $p(y \mid \cdot)$, an uncertainty measure, and region-level explanations.

Constraints.

- **BOLD-only data:** no T1-weighted image for any subject, so no anatomical registration or tissue segmentation is possible.
- **No slice-timing metadata** in any of the 175 JSON sidecars or in the raw DICOM headers. Nothing may be inferred or fabricated.
- **Heterogeneous TR:** most scans use TR ≈ 3.0 s, one uses 2.25 s and eight use ≈ 6.02 s. The last group is incompatible with the fixed 0.01–0.10 Hz band.
- **Small, imbalanced cohort:** 127 subjects, with as few as 14 in one class. One group (LMCI) has no clinical cohort table yet.
- **Compute:** a single workstation under WSL2. Processing is restricted to one worker and one acquisition at a time.

Success criteria.

- **Data level (this review):** every acquisition accounted for, with no fabricated metadata, no NaN/Inf values, and biomarkers that agree exactly with their documented definitions when recomputed.
- **Model level (final review):** under repeated, stratified, subject-grouped cross-validation, report balanced accuracy, mean absolute stage error, quadratic weighted Cohen's κ and expected calibration error, compared with non-ordinal and non-comorbidity baselines.

4. Methodology

The workflow has two halves. The *data foundation* (Modules 1–4) turns raw ADNI acquisitions into validated, model-ready features. The *modelling* half (Module 5) learns the staging model. Module 6 (testing and evaluation) runs through both halves.

4.1 Proposed Workflow

- Step 1:** Audit the raw ADNI data, recover missing scans from DICOM, document exclusions and organise everything in a BIDS-style layout.
- Step 2:** Preprocess every eligible BOLD acquisition with a single frozen pipeline and compute per-acquisition QC.
- Step 3:** Align the Brainnetome-246 atlas to each acquisition and extract ROI time series, ALFF, ReHo, FC and DC.
- Step 4:** Validate every biomarker independently and normalise within subject.
- Step 5:** Build subject-level, stratified, grouped cross-validation folds and fit cross-subject scaling on training folds only.
- Step 6:** Train the multi-attention transformer with meta-learning, fuse comorbidity information, and stage with a Bayesian ordinal head.
- Step 7:** Evaluate with ordinal and calibration metrics, and explain predictions.

4.2 Architectural Design for Proposed System

Figure 4.1 shows the complete system. Solid blocks are implemented and verified. Dashed blocks are the modelling stages still to be implemented.

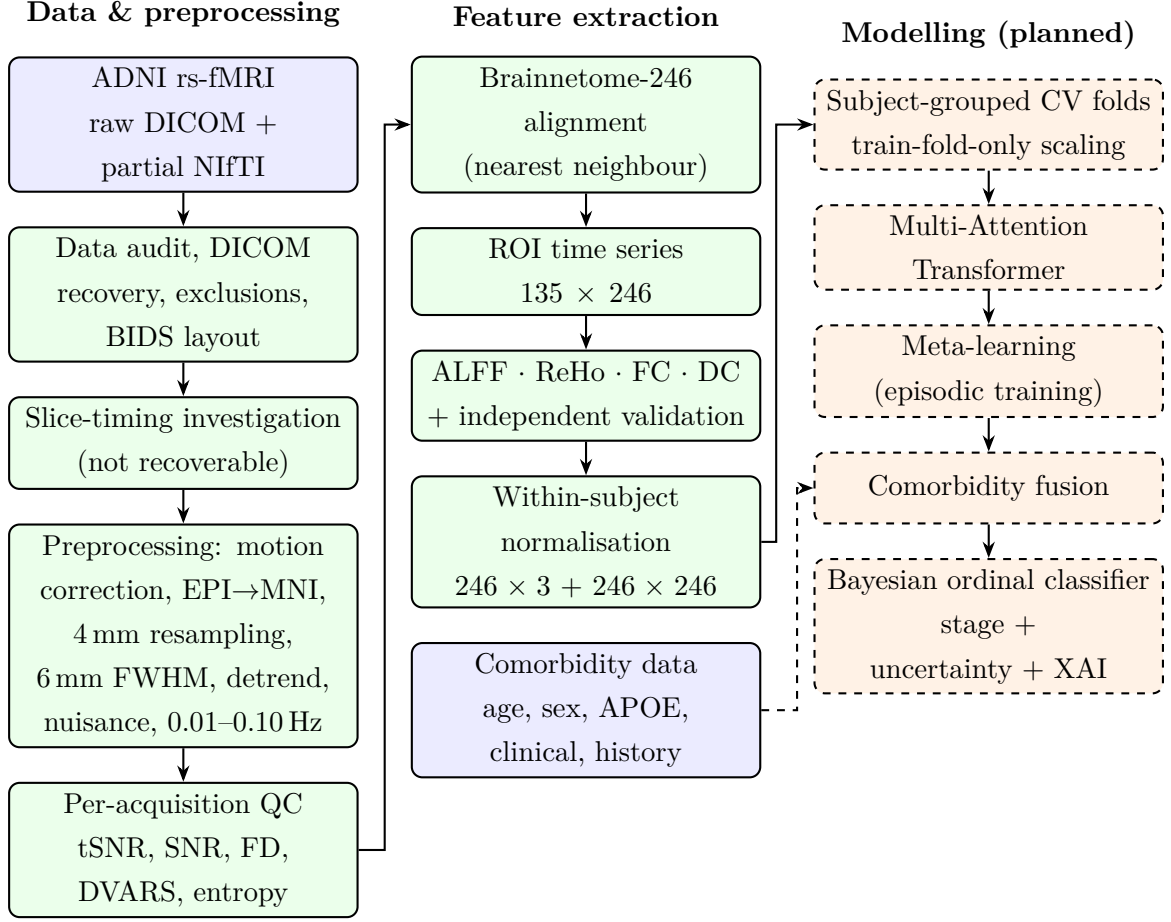


Figure 4.1: Architecture of the proposed system. Solid boxes: implemented and verified for all 165 acquisitions. Dashed boxes: planned modelling stages.

4.3 Modules

For each module the purpose, inputs, processing, outputs and dependencies are given below. Implementation detail follows in Chapter 5.

4.3.1 Module 1: Data Acquisition, Audit and Curation

<i>Purpose</i>	Produce a complete, documented inventory of usable BOLD acquisitions before any processing time is spent.
<i>Inputs</i>	ADNI raw DICOM archives, pre-converted NIfTI files and JSON sidecars for six groups; ADNI image-collection and APOE tables.
<i>Processing</i>	Inventory audit; recovery of NIfTI-missing records from raw DICOM; duplicate/orphan/MD5 hash-collision audit; exclusion register; BIDS-style organisation with <code>participants.tsv</code> as the single source of group labels; slice-timing investigation over JSON and DICOM headers.

Outputs 173 eligible acquisitions; `audit/excluded_runs.csv`; audit reports under `audit/`.

Dependencies

dcm2niix (DICOM to NIfTI), Python (pandas, pydicom).

4.3.2 Module 2: rs-fMRI Preprocessing

Purpose Remove non-neural signal and bring every scan into a common space, using one frozen pipeline for all groups.

Inputs Eligible 4-D BOLD acquisitions.

Processing Input validation $\rightarrow$ discard 5 volumes $\rightarrow$ motion correction $\rightarrow$ direct EPI-to-MNI registration $\rightarrow$ 4 mm resampling $\rightarrow$ 6 mm smoothing $\rightarrow$ linear detrending $\rightarrow$ nuisance regression $\rightarrow$ 0.01–0.10 Hz band-pass, followed by QC.

Outputs Preprocessed BOLD ($46 \times 55 \times 46 \times 135$, 4 mm), brain mask, motion parameters, nuisance design matrix, provenance JSON and a QC folder per acquisition.

Dependencies

FreeSurfer 7.4.1 FS-FAST, ANTsPy, Nilearn, TemplateFlow, SciPy.

4.3.3 Module 3: Parcellation and Biomarker Extraction

Purpose Summarise each scan as interpretable regional and network biomarkers on a fine-grained, connectivity-based atlas [11].

Inputs Preprocessed BOLD, Brainnetome-246 atlas and look-up table.

Processing Lossless atlas reorientation and nearest-neighbour alignment; ROI time series; voxel-wise ALFF and ReHo aggregated to ROIs; ROI-level FC and DC; independent recomputation checks.

Outputs Per acquisition: ROI time series (135×246), ALFF, ReHo, DC (246 each), FC (246×246), QC and metadata.

Dependencies

Module 2; NumPy, SciPy, NiBabel, Nilearn.

4.3.4 Module 4: Feature Normalisation and Model-Ready Representation

<i>Purpose</i>	Remove scanner- and subject-specific scaling without letting any information cross between subjects.
<i>Inputs</i>	Raw biomarkers from Module 3.
<i>Processing</i>	Within-subject transforms (mALFF, mReHo, DC _z , Fisher- <i>z</i> FC); redundancy analysis (FC strength removed).
<i>Outputs</i>	$\mathbf{R} \in \mathbb{R}^{246 \times 3}$ and $\mathbf{Z} \in \mathbb{R}^{246 \times 246}$ per acquisition, written to a separate output tree.
<i>Dependencies</i>	Module 3.

4.3.5 Module 5: Comorbidity-Aware Meta-Bayesian Staging Model (planned)

<i>Purpose</i>	Map imaging features and comorbidity information to an ordinal stage with calibrated uncertainty.
<i>Inputs</i>	$\mathbf{R}$, $\mathbf{Z}$ and $\mathbf{c}$ for the subjects in a training fold.
<i>Processing</i>	(i) <i>Multi-attention transformer</i> [12]: each of the 246 regions is a token embedded from its row of $\mathbf{R}$ plus a learned region embedding. Several attention heads with different inductive biases are combined, for example plain self-attention and attention whose logits are biased by $\mathbf{Z}$ so that the measured connectivity steers information flow. (ii) <i>Meta-learning</i> [13]: episodic training over small tasks (for example per acquisition site or per adjacent-stage pair) to learn representations that adapt with few examples, which suits classes as small as 14 subjects. (iii) <i>Comorbidity fusion</i>: $\mathbf{c}$ is embedded and fused with the imaging representation through gating or cross-attention. (iv) <i>Bayesian ordinal classifier</i>: a cumulative-link head [14],

$$P(y \leq k \mid \mathbf{h}) = \sigma(\theta_k - g(\mathbf{h})), \quad \theta_0 < \theta_1 < \dots < \theta_4, \quad (4.1)$$

with priors on weights and thresholds. The posterior is approximated (for example by variational inference or Monte Carlo dropout [15]), giving a predictive distribution and its uncertainty.

Outputs $p(y \mid \mathbf{R}, \mathbf{Z}, \mathbf{c})$, predicted stage, uncertainty, attention- and attribution-based regional explanations.

Dependencies

Modules 4 and 6; PyTorch (planned).

4.3.6 Module 6: Testing and Evaluation

Purpose Make every stage verifiable and every model result unbiased.

Data-level testing (done)

Input validation, per-acquisition QC, a pilot-reproduction gate before scale-up, independent re-reading and recomputation of biomarkers, normalisation checks and dataset-wide audits.

Model-level evaluation (planned)

Repeated, stratified, *subject-grouped* k -fold cross-validation (no subject in both train and test); all cross-subject statistics fitted on the training fold only [16, 17]; metrics: balanced accuracy, mean absolute stage error, quadratic weighted κ , expected calibration error; ablations removing ordinal structure, comorbidity fusion and meta-learning.

5. Implementation

5.1 Environment and Engineering Practices

All processing runs on a Windows 11 workstation under WSL2 (Ubuntu). Table 5.1 lists the software versions recorded in the provenance files.

Table 5.1: Software environment (from provenance JSON sidecars)

Component	Version	Component	Version
Python	3.12.3	Pandas	3.0.5
NumPy	2.3.5	Matplotlib	3.11.2
SciPy	1.15.3	ANTsPy (antspyx)	0.6.3
NiBabel	5.4.2	TemplateFlow	25.1.2
Nilearn	0.14.1	FreeSurfer (FS-FAST)	7.4.1

Three engineering decisions made long runs on this hardware reliable:

- **Single worker, pinned threads.** `n_procs=1`; ANTs, ITK, OpenMP, OpenBLAS and MKL thread counts are pinned to 1. Acquisitions are processed one at a time, with scratch clean-up and garbage collection after each.
- **Validated checkpoint/resume.** `PROCESSING_CHECKPOINT.json` records completed, failed and remaining acquisitions per group after *every* acquisition. On resume, outputs are re-opened and checked (4-D BOLD loadable, 3-D mask, motion/design/JSON readable) rather than trusted from the checkpoint.
- **Crash-aware retry wrappers.** Host-memory pressure caused recurring WSL2 VM crashes. The `scripts/setup/retry_*.sh` wrappers tell a VM crash (no completion marker, no traceback) apart from a genuine Python error (traceback present), restart WSL and resume automatically. The biomarker stage also stages each acquisition on the WSL-native filesystem, because sustained NumPy I/O on the Windows mount raised `Errno 5` failures.

5.2 Module 1: Data Audit and Curation

Initial audit. The first inventory found 131 subjects but only 72 valid NIfTI files. 103 records had JSON metadata without a matching NIfTI, 35 subjects had multiple

acquisitions and **no subject had a T1w image**.

Recovery. Missing records were matched against the raw DICOM archives and converted (Table 5.2). For SMC, 20 of the group’s 32 final acquisitions (62.5%) exist only because of this step. An AD re-audit also overturned an earlier note that 6 AD scans were unrecoverable (R3).

Table 5.2: DICOM recovery of acquisitions with missing NIfTI files

Group	JSON records	NIfTI present	Missing	Converted	Failures
MCI	35	26	9	9	0
SMC	32	12	20	20	0

Duplicates and exclusions. A dataset-wide audit of 175 JSON and 175 NIfTI files found no orphans and one MD5 hash collision: LMCI subject 006_S_4363 had the same series exported twice. Two acquisitions were excluded before preprocessing (Table 5.3). Their source files are retained, and every batch script consults the exclusion register.

Table 5.3: Documented pre-processing exclusions (`audit/excluded_runs.csv`)

Group	Acquisition	Reason
CN	sub-012S4026 / ses-01 / run-01	Only 7 volumes (others have 140); truncation present in the source DICOM export
LMCI	sub-006S4363 / ses-01 / run-02	Byte-identical duplicate of run-01 (MD5 <code>e7a650bf...</code>)

Slice-timing investigation. None of the 175 JSON sidecars contains a `SliceTiming` field. Representative raw DICOM headers from all 17 distinct protocol signatures were then inspected element by element, including the Philips private groups. `AcquisitionTime` and `TriggerTime` are identical for every slice within a volume, so timing is known only per volume. The private tag (2001,100A) gives a slice’s *spatial* index, not its temporal order, and it was deliberately not used to guess one. No named multiband tag exists. The outcome is `SLICE_TIMING_NOT_RECOVERABLE`, so slice-timing correction is not performed. Every provenance file records `"slice_timing_status": "NOT_PERFORMED"`. The audit also found that, for the $TR \approx 6.02$ s protocol, the DICOM `RepetitionTime` tag

under-reports the effective TR by about $2\times$. The pipeline therefore takes TR from the BIDS sidecar and NIfTI header.

5.3 Module 2: Preprocessing Pipeline

The frozen pipeline is `pipeline/fsfast_production_4mm6mm.py` (function `process_one_acquisition`). Table 5.4 summarises it.

Table 5.4: Preprocessing steps and parameters

Step	Tool	Parameters / notes
Input validation	NumPy / NiBabel	4-D; no NaN/Inf; finite affine with $ \det > 10^{-9}$; $TR > 0$; non-empty
Volume removal	array slicing	first 5 volumes discarded ($140 \rightarrow 135$)
Slice timing	–	not performed (metadata not recoverable)
Motion correction	FS-FAST <code>mc-afni2</code> (AFNI <code>3dvolreg</code>) [18, 19]	rigid 6-DOF, reference = first retained frame; native <code>.mcdat</code> kept
Normalisation	ANTsPy SyN [20]	mean EPI (Otsu skull-strip) $\rightarrow$ MNI152NLin6Asym 2 mm [21]; single-step 4-D warp
Resampling	Nilearn [22]	4 mm isotropic, continuous interpolation
Smoothing	Nilearn	Gaussian, 6 mm FWHM
Detrending	OLS	linear only, temporal mean restored
Nuisance regression	OLS	Friston-24 + WM + [CSF] + global signal + intercept; reliability-gated
Band-pass	SciPy	Butterworth order 2, <code>filtfilt</code> , 0.01–0.10 Hz; Nyquist checked

Motion and framewise displacement. From the six rigid-body parameters (translations d_x, d_y, d_z in mm; rotations α, β, γ in radians), framewise displacement follows Power et al. [23]:

$$FD_t = |\Delta d_{x,t}| + |\Delta d_{y,t}| + |\Delta d_{z,t}| + r (|\Delta \alpha_t| + |\Delta \beta_t| + |\Delta \gamma_t|), \quad r = 50 \text{ mm.} \quad (5.1)$$

High-motion acquisitions are flagged `WARN`. By the frozen policy they are never excluded automatically.

Registration without T1w. With no anatomical image, the temporal mean of the motion-corrected series is registered directly to the MNI152NLin6Asym template (rigid + affine + SyN). The whole 4-D series is warped in one interpolation step to limit blurring.

Nuisance regression. The Friston-24 expansion [24] of the motion parameters $\mathbf{m}_t \in \mathbb{R}^6$ is

$$\mathbf{F}_t = [\mathbf{m}_t, \mathbf{m}_t - \mathbf{m}_{t-1}, \mathbf{m}_t^2, (\mathbf{m}_t - \mathbf{m}_{t-1})^2] \in \mathbb{R}^{24}. \quad (5.2)$$

Subject-specific tissue segmentation is impossible without T1w, so WM and CSF signals come from population priors (MNI152NLin2009cAsym probability maps; WM $p \geq 0.95$ with one erosion, CSF $p \geq 0.90$). A tissue regressor is kept only if its mask has at least 20 voxels *and* its correlation with the global signal satisfies $|r| < 0.98$. With design matrix $\mathbf{X}$ and voxel signal $\mathbf{y}$, the cleaned signal is

$$\tilde{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} + \bar{y}, \quad \hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2. \quad (5.3)$$

In the production run, the WM regressor passed the gate in all 165 acquisitions, and the CSF regressor *failed it in all 165*: the thin ventricular prior leaves very little mask once resampled to 4 mm and intersected with the brain mask. Every design matrix therefore has **27 columns**: intercept, Friston-24, WM and global signal. The rejection is logged per acquisition ("**CSF_status**": "**NOT PERFORMED**"). No substitute regressor was invented. Global-signal regression is a debated choice in the literature [26], and it is recorded explicitly.

Band-pass and Nyquist validation. TR is read from each header. Before filtering, the pipeline checks $0 < 0.01 < 0.10 < f_N = 1/(2 \text{TR})$. For the 8 acquisitions acquired at $\text{TR} = 6.02$ s, $f_N = 0.083$ Hz, which is below the 0.10 Hz cut-off. These acquisitions were marked FAIL with the exact reason. The band was never silently changed.

QC metrics. For each acquisition the pipeline stores: tSNR ($\text{mean}|x_v|/\text{std}(x_v)$ on the smoothed, pre-regression data); native-space SNR (brain mean / background standard deviation); FD; raw-intensity DVARS, $\text{DVARS}_t = \sqrt{\langle (x_{v,t} - x_{v,t-1})^2 \rangle_v}$; and spatial and temporal Shannon entropy. CNR is recorded as **NOT_RELIABLY_COMPUTABLE** because a tissue-contrast definition needs T1w. Raw-intensity DVARS is never compared with literature thresholds, which assume standardised DVARS.

5.4 Module 3: Parcellation and Biomarker Extraction

Implemented in `scripts/biomarkers/biomarker_dataset_pipeline.py` (modes `discover`, `verify-pilot`, `run`, `audit`).

Atlas alignment. The 2 mm Brainnetome-246 atlas [11] is stored in LAS orientation. It is reoriented to RAS by an exact array flip (no interpolation) and resampled onto each acquisition’s own BOLD grid with *nearest-neighbour* interpolation, because labels are categorical. BOLD itself uses continuous interpolation. Labels 1–246 were enumerated and confirmed present.

ROI time series. Each ROI’s signal is the mean BOLD over its voxels at each time point. It is computed by direct index masking on the verified identical grid, which avoids any hidden second resampling. The orientation is fixed as $\text{time} \times \text{ROI}$, (135, 246).

ALFF. ALFF is computed per voxel v from the mean-centred series and averaged within each ROI [7]:

$$\text{ALFF}_v = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} |\mathcal{F}\{x_v - \bar{x}_v\}(f_k)|, \quad \mathcal{K} = \{k : 0.01 \leq f_k \leq 0.10 \text{ Hz}\}. \quad (5.4)$$

ReHo. ReHo is Kendall’s coefficient of concordance over the 27-voxel ($3 \times 3 \times 3$) neighbourhood [8]. Near the mask boundary, only in-mask neighbours are used and K is that voxel’s actual neighbour count:

$$W = \frac{\sum_{i=1}^n R_i^2 - n\bar{R}^2}{\frac{1}{12}K^2(n^3 - n)}, \quad (5.5)$$

where R_i is the sum of the K ranks at time point i and $n = 135$. The dataset-scale version precomputes ranks once and accumulates neighbour rank sums by shifted array addition. This cut ReHo time from about 68 s to about 3 s per acquisition and stayed bit-identical to the reference loop.

FC, DC and FC strength. FC is the full Pearson correlation matrix of the ROI series: signed, unthresholded, diagonal kept. Weighted, signed degree centrality [9]

and regional FC strength are

$$\text{DC}_i = \sum_{j \neq i} \text{FC}_{ij}, \quad \text{FCS}_i = \frac{1}{245} \sum_{j \neq i} \text{FC}_{ij}. \quad (5.6)$$

Negative DC values are an expected consequence of keeping negative correlations.

Pilot-reproduction gate. Before any other acquisition was processed, the dataset-scale code had to reproduce the already-validated pilot (`AD/sub-019S4549/ses-01/run-01`) in a scratch directory (Table 5.5). The pilot directory was then written in no-clobber mode, so its original files are untouched.

Table 5.5: Pilot-reproduction gate: maximum absolute difference against the validated pilot

Quantity	Shape	Max. abs. difference
ALFF	(246,)	0.000 (bit-identical)
ReHo	(246,)	0.000 (bit-identical)
ROI time series	(135, 246)	0.000 (bit-identical)
Aligned atlas labels / voxel counts	–	0.000 (bit-identical)
FC	(246, 246)	3.3×10^{-16}
Degree centrality	(246,)	1.8×10^{-15}
FC strength	(246,)	9.9×10^{-17}

5.5 Module 4: Within-Subject Normalisation

Implemented in `scripts/biomarkers/normalize_within_subject.py`. Raw ALFF is in FFT amplitude units that scale with each scan’s raw intensity. Mean ALFF ranges from 850 to 215,172 across acquisitions, a factor of about 250. Every statistic in this stage comes from *one acquisition’s own 246 values*. No cross-subject statistic or label is used, so the stage cannot leak information between training and test subjects. Cross-subject scaling is deliberately left to the modelling stage, where it will be fitted on training folds only.

Table 5.6: Within-subject transforms

Output	Transform	Rationale
mALFF	$\text{ALFF}_i / \overline{\text{ALFF}}$	removes raw-intensity scaling; ratio form as in the literature
mReHo	$\text{ReHo}_i / \overline{\text{ReHo}}$	removes the subject’s global coherence offset
DC_z	$(\text{DC}_i - \overline{\text{DC}}) / \sigma_{\text{DC}}$	z-score because DC is signed (σ with ddof=0)
Fisher-z FC	$\text{artanh}(\text{clip}(r_{ij}, \pm 0.999999)), \text{diag} = 0$	variance-stabilises correlations; diagonal zeroed <i>before</i> artanh so no ∞ is ever produced
FC strength	<i>excluded</i>	FCS = DC/245, perfectly collinear; raw files retained

Outputs go to a separate tree, `derivatives/biomarkers_normalized/`. The script raises an error if a write path ever points inside a raw `biomarkers/` directory. Per acquisition, the model-ready representation is the regional matrix $\mathbf{R} \in \mathbb{R}^{246 \times 3}$ (mALFF, mReHo, DC_z) plus the connectivity matrix $\mathbf{Z} \in \mathbb{R}^{246 \times 246}$.

5.6 Module 5: Status of the Modelling Stages

The modelling stages are not yet implemented. Their prerequisites are now in place, and the next steps are:

1. Obtain and clean the comorbidity tables. APOE genotype from the ADNI APOE results table is available for 126 of the 127 subjects. The clinical cohort tables (CDR, MMSE, GDS, medical history) are present for five groups but missing for LMCI. Because this gap follows group lines, an availability flag would act as a proxy for the LMCI label, so it must not be used as a feature.
2. Build a subject-level cohort index and repeated, stratified, subject-grouped cross-validation folds.
3. Fit cross-subject scaling on training folds only, then implement and train the transformer, meta-learning, fusion and ordinal head.

5.7 Output Organisation

```
derivatives/
  fsfast/<GROUP>/sub-<ID>/ses-<NN>/run-<NN>/
    *_desc-preproc_bold.nii.gz / .json    final BOLD + provenance
    *_motion_parameters.mcdat             native mc-afni2 output
    brain_mask.nii.gz, design_matrix.txt
    qc/                                   qc_metrics.csv, fd/dvars values, QC figures
    biomarkers/ roi_timeseries, alff, reho, fc_matrix,
                  degree centrality, fc_strength, QC + metadata
  fsfast/FINAL_6GROUP_AUDIT/             dataset-wide preprocessing audit
  fsfast/BIOMARKER_DATASET_AUDIT/        biomarker manifest, QC, pilot gate
  biomarkers_normalized/<GROUP>/.../
    malff, mreho, dc_z, fc_fisher_z,
    regional_biomarkers_normalized (246 x 3), metadata
```


6. Results and Discussion

6.1 Dataset Accounting

Figure 6.1 traces every acquisition from the raw dataset to the model-ready features. Table 6.1 gives the per-group breakdown and demographics.

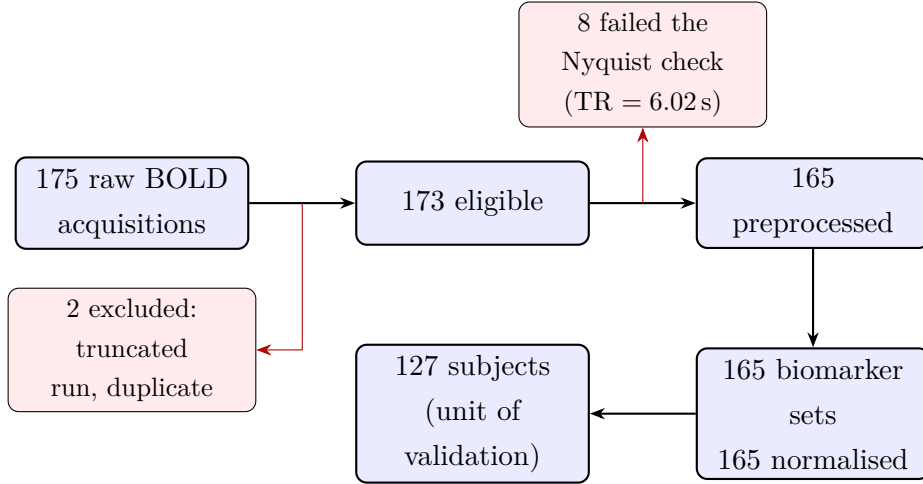


Figure 6.1: Acquisition flow from raw ADNI data to model-ready features

Table 6.1: Per-group cohort summary (processed subjects only). Age and sex are taken from the ADNI image-collection tables, at the earliest rs-fMRI scan listed for each subject. APOE $\epsilon 4+$ counts subjects carrying at least one $\epsilon 4$ allele.

	CN	SMC	EMCI	MCI	LMCI	AD
Eligible acquisitions	29	32	30	35	22	25
Preprocessed / biomarkers	27	31	29	32	21	25
Subjects (processed)	26	21	26	14	20	20
Scans per subject	1.04	1.48	1.12	2.29	1.05	1.25
Age (mean $\pm$ SD)	76.8 $\pm$ 6.4	72.0 $\pm$ 5.4	70.4 $\pm$ 7.1	71.8 $\pm$ 8.1	[n/a]	75.3 $\pm$ 7.7
Sex (F/M)	14/12	12/9	16/10	5/9	[n/a]	11/9
APOE $\epsilon 4+$ / genotyped	8/26	8/20	17/26	5/14	7/20	19/20

Two points stand out. First, sessions inflate the MCI class: it is the largest by scans but the smallest by subjects. Splitting by scan would put the same brain in both training and test sets, so validation must be subject-grouped. Second, APOE $\epsilon 4$ carriage is 31% in CN and 95% in AD, consistent with its known role as the strongest common genetic risk factor [10]. The intermediate groups do not follow a clean trend

(65% in EMCI, 36% in MCI, 35% in LMCI), which is unsurprising with 14–26 subjects per group. APOE is therefore a plausible input for the comorbidity branch, not a stage marker on its own. The LMCI demographics are not available until its cohort table is obtained.

6.2 Preprocessing Quality Control

Table 6.2 and Figure 6.2 summarise QC across the 165 acquisitions. Medians are reported for tSNR because one CN acquisition (**sub-018S4313**, ses-01) has near-zero-variance voxels at the mask edge. They push its mean tSNR to 3.2×10^5 , and the group *mean* in the raw audit table to 12,064. This was investigated and documented rather than silently removed.

Table 6.2: Preprocessing QC by group (165 acquisitions)

	CN	SMC	EMCI	MCI	LMCI	AD
Acquisitions	27	31	29	32	21	25
tSNR, median	132.8	123.7	133.2	139.2	121.4	118.8
SNR, mean	13.03	13.15	13.76	12.94	12.99	13.49
FD (mm), mean	0.379	0.375	0.355	0.321	0.322	0.439
Volumes with FD > 0.5 mm (%)	19.8	23.8	20.4	16.4	16.5	27.7
Spatial entropy (bits)	7.09	7.15	7.15	7.17	7.18	7.22
Temporal entropy (bits)	3.55	3.57	3.53	3.56	3.51	3.53
WM / CSF regressor used	27/0	31/0	29/0	32/0	21/0	25/0

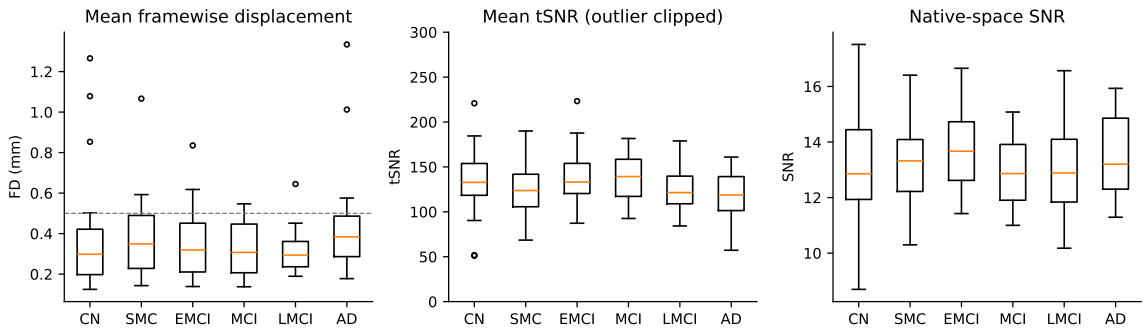


Figure 6.2: Per-acquisition QC by group. Dashed line: 0.5 mm FD reference. The tSNR axis is clipped at 300 to show the bulk of the distribution.

Discussion. QC is broadly comparable across groups. Differences in signal quality are modest and do not follow the stage order (median tSNR 119–139, mean SNR

12.9–13.8). This matters because a quality difference between classes could be learnt as a spurious “biomarker”. AD does have both the lowest median tSNR and the highest motion (mean FD 0.44 mm, 27.7% of volumes above 0.5 mm), so both will be checked as confounds. 155 of the 165 acquisitions carry a WARN flag, every one of them for FD. Under the frozen policy these are flagged, not excluded. Because motion is known to bias connectivity estimates [23, 25], mean FD will be used as a covariate and confound check in the modelling stage. All 165 acquisitions are 4-D, on the identical 4 mm grid, with valid masks and no NaN/Inf. Source checksums were unchanged before and after processing in 165/165.

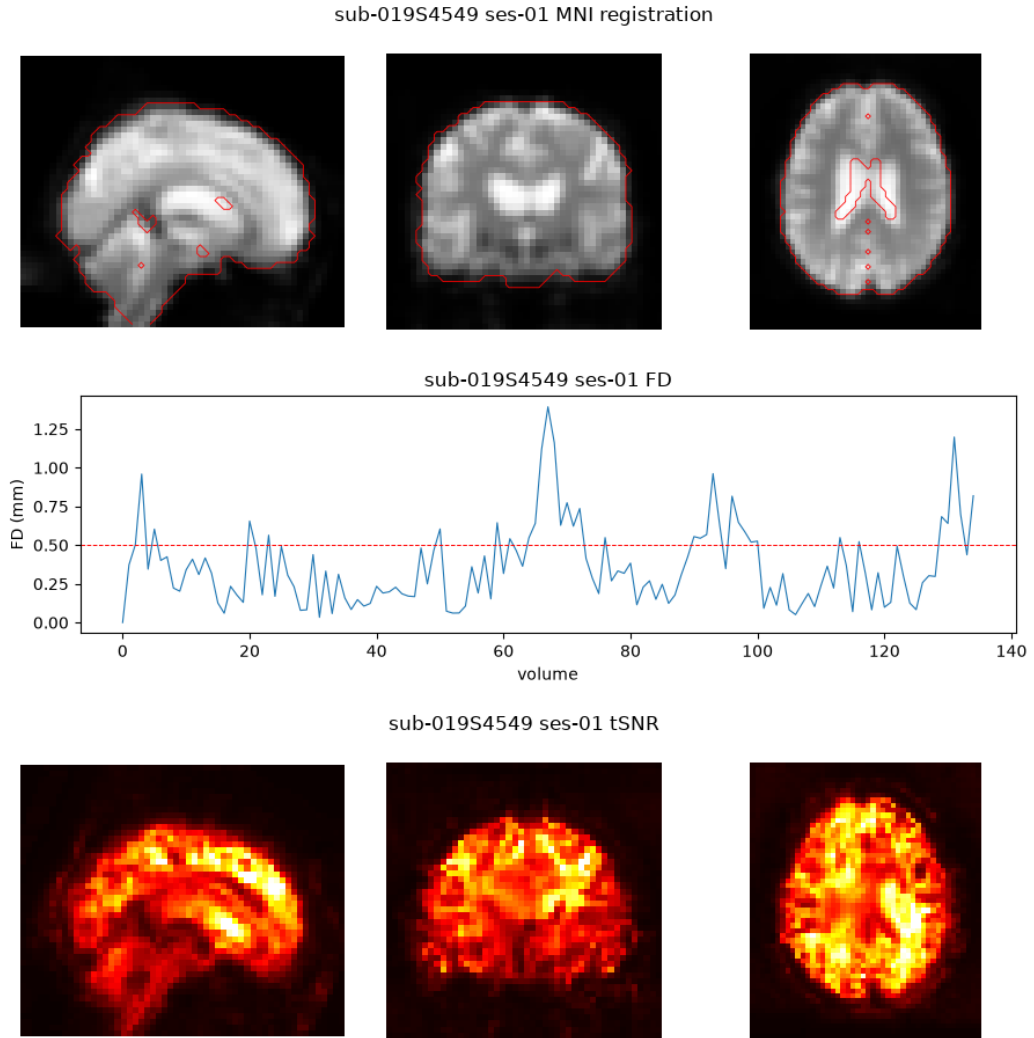


Figure 6.3: Pipeline QC output for the pilot acquisition (AD, sub-019S4549, ses-01): MNI registration overlay (top), framewise displacement (middle) and tSNR map (bottom).

6.3 Parcellation and Biomarker Validation

Table 6.3 gives the dataset-level verification over all 165 acquisitions. The checks re-read the saved files and recompute the derived quantities independently of the generation code.

Table 6.3: Dataset-level biomarker verification (165 acquisitions)

Check	Result
Acquisitions with all 246 ROIs present / any zero-voxel ROI	165/165 / 0
ROI voxel count (min / median / max)	10 / 68 / 186
ROI time series shape (135, 246)	165/165
Non-finite values; zero-variance ROIs	0; 0
ALFF non-finite or negative values	0
ReHo values outside [0, 1]; observed range	0; 0.235–0.823
FC max diagonal error / max asymmetry	2.2×10^{-16} / 2.2×10^{-16}
FC negative off-diagonal share (mean, range)	50.1% (45.6–53.1%)
DC recomputed from stored FC, max difference	0.000
FC strength recomputed, max difference	6.9×10^{-18}
Acquisitions passing all consistency checks; failures	165/165; 0

The DC result is the strongest evidence here. It shows that the files are well formed and also that the stored values are exactly what the documented definition produces, in every acquisition. About half of all off-diagonal correlations are negative. This is expected after global-signal regression [26], and it is why DC and FC are kept signed.

6.4 Normalisation Results

Figure 6.4 shows why within-subject normalisation is needed. Raw mean ALFF spans more than two orders of magnitude between acquisitions and mostly reflects intensity scaling, while ReHo stays in a narrow band. After normalisation, mean mALFF and mean mReHo equal 1.000000 and $\text{std}(\text{DC}_z)$ equals 1.000000 in all 165 acquisitions. Mean DC_z is at most 1.6×10^{-16} . The Fisher- z FC is symmetric (max asymmetry 5.3×10^{-15}), its diagonal is exactly 0 in 165/165 and its range is -2.27 to 2.59 . The ± 0.999999 clip was *never* triggered, and no input or output contains NaN or Inf.

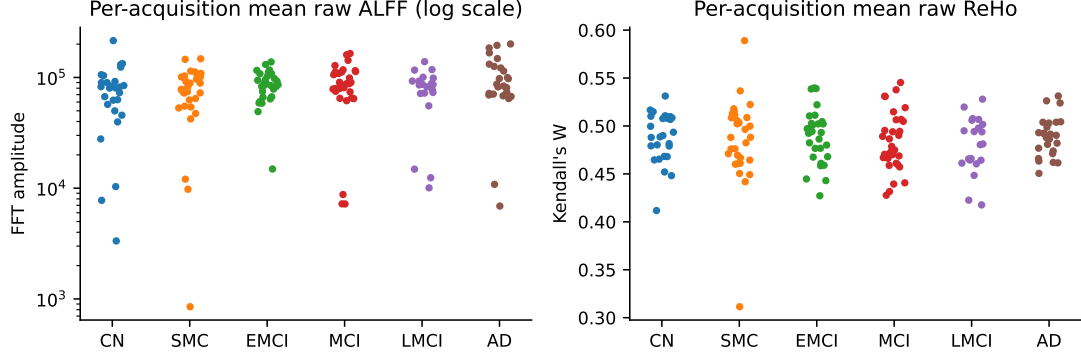


Figure 6.4: Per-acquisition mean raw ALFF (log scale) and raw ReHo by group. ALFF varies by a factor of about 250 because it scales with raw BOLD intensity.

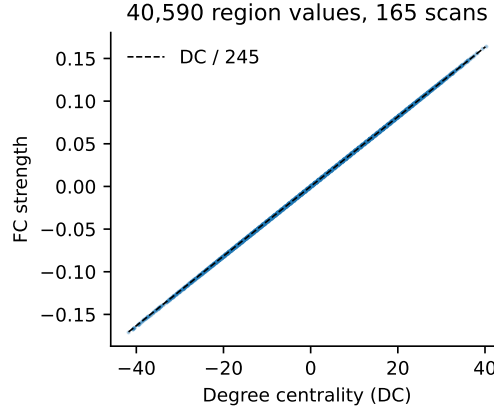


Figure 6.5: DC against FC strength for all 40,590 region values. Every point lies on $FCS = DC/245$ ($r = 1.0$), so FC strength is excluded as redundant.

6.5 Descriptive Group-Level Connectivity

Figure 6.6 shows the group-mean Fisher- z connectivity, averaged first within each subject and then across subjects so that multi-session subjects are not over-weighted. All six groups show the same broad structure: strong positive coupling within lobes (the diagonal blocks), a distinct subcortical block, and weak or negative coupling between several lobes. These maps are *descriptive only*. No statistical test has been run, and group means do not show whether individuals can be told apart. That question is for the planned model, under subject-grouped validation.

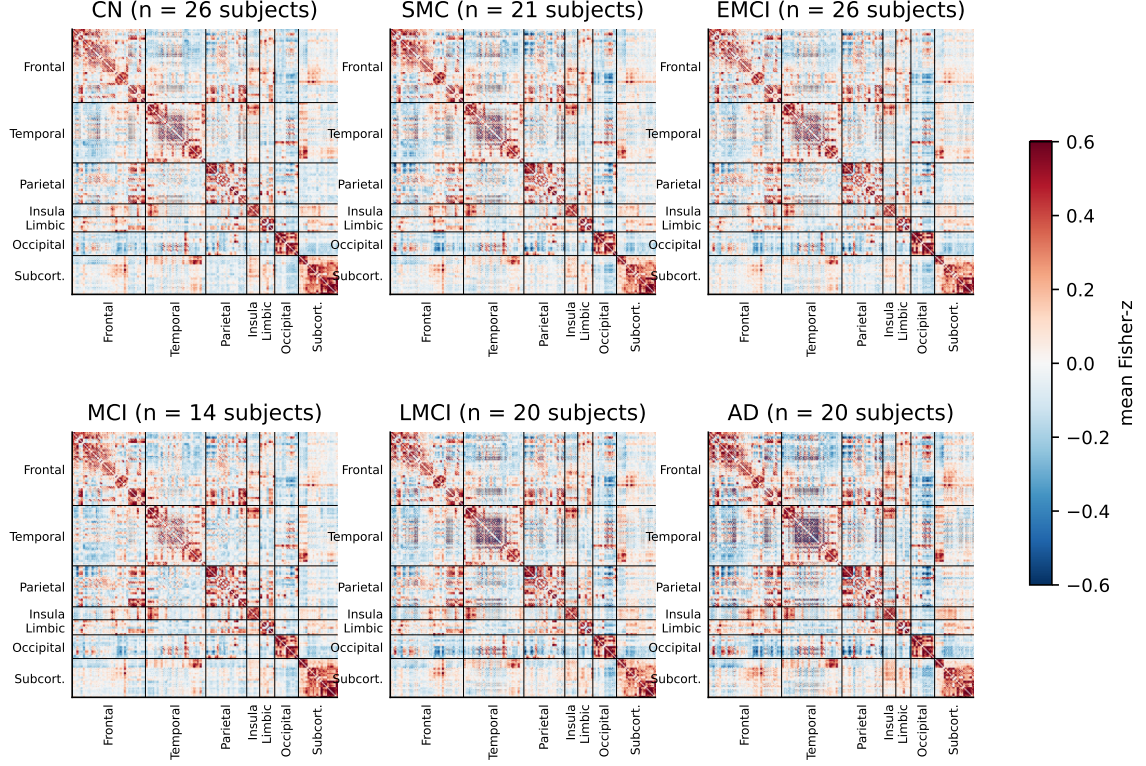


Figure 6.6: Group-mean Fisher- z functional connectivity on the Brainnetome-246 atlas, ordered by lobe (subject-level averaging).

6.6 Comparison with Existing Work

Table 6.4 compares the design choices of this pipeline with those of conventional T1w-based rs-fMRI pipelines such as DPABI [27]. The main differences come from the dataset constraints. Where a conventional pipeline would rely on missing information, this project substitutes a documented alternative or records the step as not performed.

Table 6.4: Design comparison with a conventional rs-fMRI pipeline

Aspect	Typical pipeline with T1w and full metadata	This project (BOLD-only ADNI subset)
Spatial normalisation	BOLD $\rightarrow$ T1w $\rightarrow$ template	direct EPI $\rightarrow$ MNI152 (ANTs SyN)
WM/CSF signals	subject-specific tissue segmentation	template priors with a per-acquisition reliability gate
Slice timing	corrected from header timing	audited as not recoverable; not performed, not guessed
Band-pass vs. TR	one fixed band for all scans	band checked against each scan's Nyquist frequency; 8 incompatible scans rejected with logged reason
Atlas labels	toolbox-internal resampling	nearest-neighbour only, coverage verified per scan
Biomarker verification	toolbox outputs used directly	exact independent recomputation, 165/165
Scaling / splitting	whole-dataset statistics and scan-level splits are known leakage sources [17]	within-subject now; cross-subject only on training folds; subject-grouped CV

Leakage between training and test data, through subject overlap or whole-dataset preprocessing statistics, is a recurring problem in AD classification studies and can inflate reported accuracy substantially [17]. Small neuroimaging samples also produce wide error bars [16]. This project avoids both by design. Every statistic computed so far is within-acquisition, and the evaluation protocol is fixed as repeated, stratified, subject-grouped cross-validation before any model is trained. Accuracy comparisons with published classifiers will be made in the final review, once the model exists.

6.7 Limitations

- No T1w image exists, so registration is EPI-to-template, WM/CSF signals come from priors (the CSF prior failed the gate in every acquisition), and CNR cannot be computed.
- Slice-timing correction was impossible. Undocumented private DICOM tags cannot be completely ruled out as carrying such information.

- 8 of 173 eligible acquisitions ($TR = 6.02$ s) are excluded unless the methodology is revisited for that protocol.
- Validation so far is mathematical and structural, not biological; diagnostic value can only be shown by the model.
- The cohort is small (127 subjects, 14 in MCI). The LMCI clinical table is missing. Hypertension and diabetes are available only as broad medical-history categories.

6.8 Plan for the Remaining Work

Figure 6.7 gives the tentative schedule up to the final review.

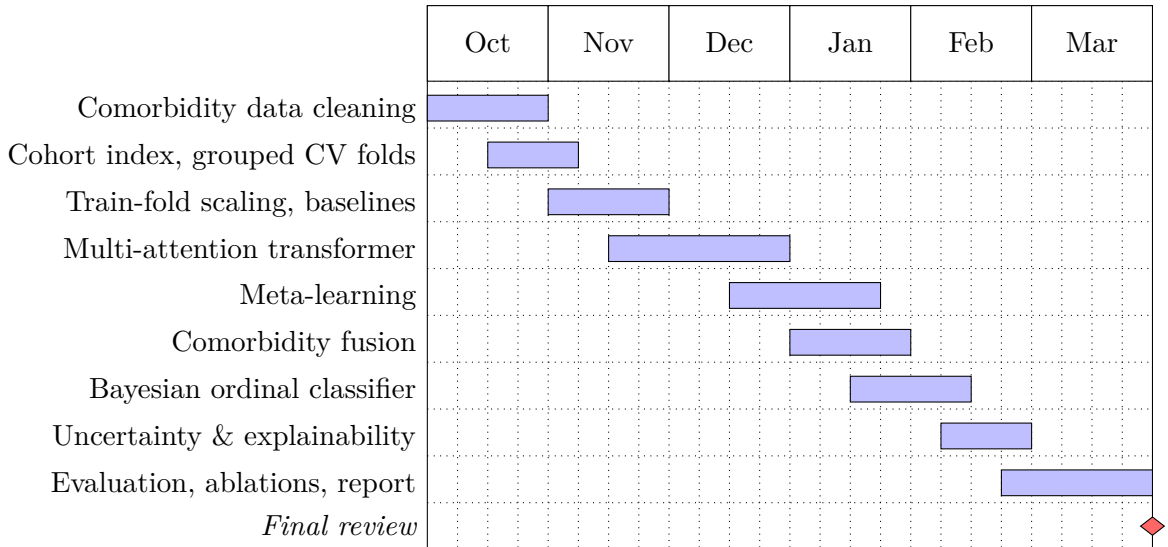


Figure 6.7: Tentative schedule for the modelling stages (weeks grouped by month)

6.9 Conclusion

The data foundation is complete. 175 ADNI rs-fMRI acquisitions have been audited, and 165 have been preprocessed, parcellated, turned into ALFF, ReHo, FC and DC biomarkers and normalised within subject, with zero processing failures after preprocessing. Every derived value has been verified independently. Each limitation of the BOLD-only dataset is handled by a documented substitution or exclusion rather than by fabrication. The work also corrected the Review-I design in two places. The regional feature matrix is 246×3 , because FC strength is redundant. The unit of validation is the subject, of which there are 127. The next phase builds

the comorbidity-aware Meta-Bayesian ordinal model on these features and evaluates it under subject-grouped cross-validation.

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